Place Design (103 errors)

[Place 30-415] IO Placement failed due to overutilization. This design contains 104 I/O ports
while the target device: 7s6 package: ftgb196, contains only 100 available user I/O. The target device has 100 usable I/O pins of which 0 are already occupied by user-locked I/Os.
To rectify this issue:
1. Ensure you are targeting the correct device and package. Select a larger device or different package if necessary.
2. Check the top-level ports of the design to ensure the correct number of ports are specified.
3. Consider design changes to reduce the number of I/Os necessary.

[Place 30-68] Instance adc_ch0_IBUF[0]_inst (IBUF) is not placed (99 more like this)

From **Spartan** to **Artix Architecture**

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	-4.599 ns	Worst Hold Slack (WHS):	0.089 ns	Worst Pulse Width Slack (WPWS):	3.950 ns
Total Negative Slack (TNS):	-43.792 ns	Total Hold Slack (THS):	0.000 ns	Total Pulse Width Negative Slack (TPWS):	0.000 ns
Number of Failing Endpoints:	10	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	1051	Total Number of Endpoints:	1051	Total Number of Endpoints:	344

Timing constraints are not met.

Applying **Pipelining** to the Packetizer Module, making the **USB-output** Asynchronous.

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	2.857 ns	Worst Hold Slack (WHS):	0.072 ns	Worst Pulse Width Slack (WPWS):	2.950 ns
Total Negative Slack (TNS):	0.000 ns	Total Hold Slack (THS):	0.000 ns	Total Pulse Width Negative Slack (TPWS):	0.000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	1068	Total Number of Endpoints:	1068	Total Number of Endpoints:	356

All user specified timing constraints are met.

Project Summary | Path 1 - daq_top_timing_summary_routed

Summary

Name	Path 1
Slack	-1.425ns
Source	u_packetizer/state_curr_reg[2]_replica_3/C (rising edge-triggered cell FDRE clocked by sys_clk {rise@0.000ns fall@5.000ns per
Destination	tx_data[5] (output port clocked by sys_clk {rise@0.000ns fall@5.000ns period=10.000ns})
Path Group	sys_clk
Path Type	Max at Slow Process Corner
Requirement	10.000ns (sys_clk rise@10.000ns - sys_clk rise@0.000ns)
Data Path Delay	5.659ns (logic 2.991ns (52.848%) route 2.669ns (47.152%))
Logic Levels	5 (LUT5=1 LUT6=3 OBUF=1)
Output Delay	2.000ns
Clock Path Skew	-3.730ns
Clock Un...rtainty	0.035ns

Source Clock Path

Delay Type	Incr (ns)	Path ...	Location	Netlist Resource(s)
(clock sys_...rise edge)	(r) 0.000	0.000		
	(r) 0.000	0.000	Site: N13	clk
net (fo=0)	0.000	0.000		clk
IBUF (Prop_ibuf_I_O)	(r) 0.850	0.850	Site: N13	clk_IBUF_inst/O

Fig. Path Delay due to Combinatorial State Machine output

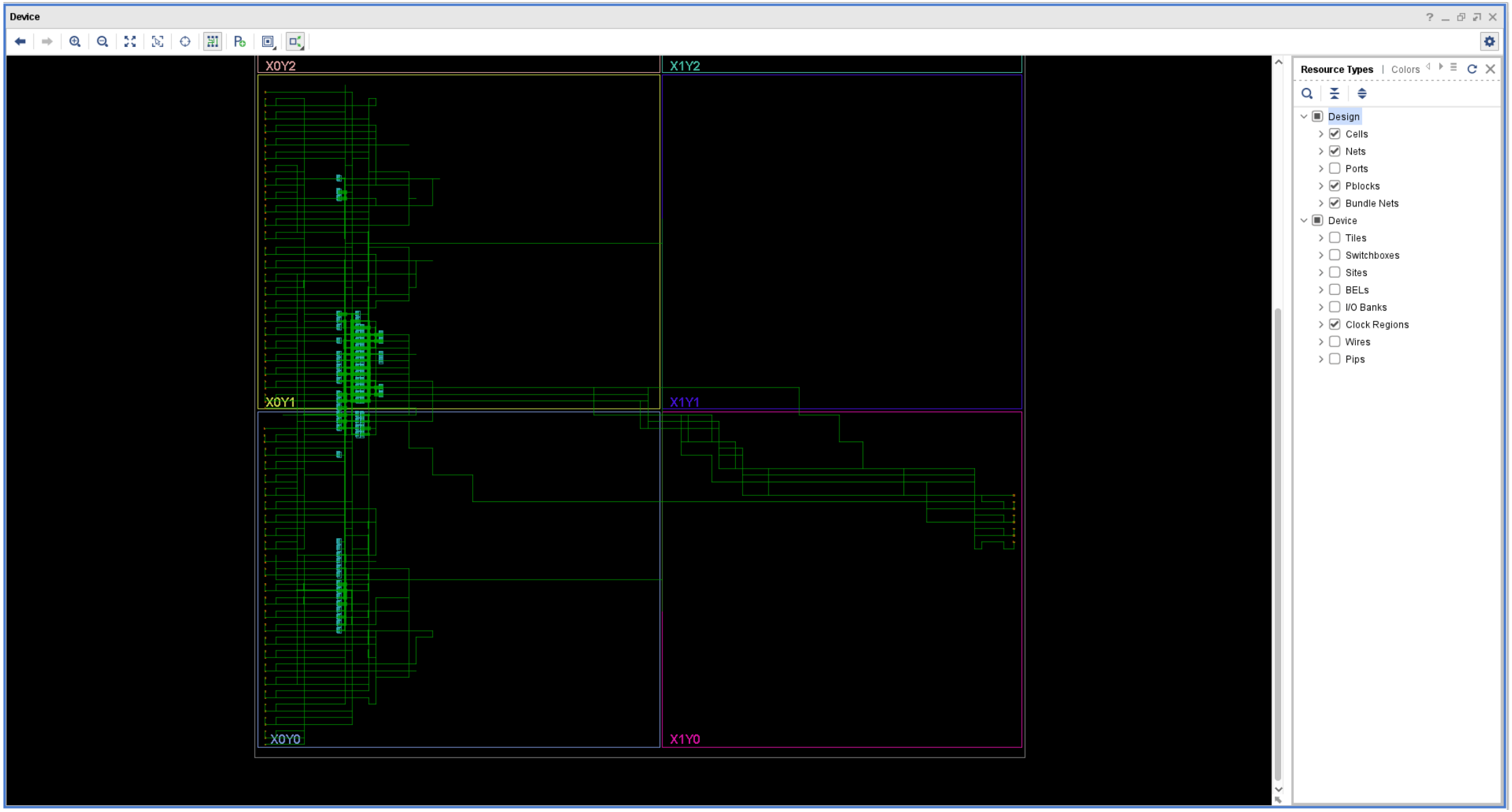

Fig. Artix-7(xc7a35tftg256-3) Implementation

Fig. Artix-7(xc7a35tftg256-3) Device Pinout (Marked in Brown)